

claude-windows / 02b9e9c9-3806-45ca-9b24-2ecc35a81efd

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\02b9e9c9-3806-45ca-9b24-2ecc35a81efd\subagents\workflows\wf_27df53c8-8e0\agent-af4238ba05656216b.jsonl`  
- SHA-256: `0c67df9182418e742f1fe49456e08a3992d9c5ebd75487c6e314d94042383ffe`  
- Source modified: `2026-06-10T16:31:15+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_27df53c8-8e0`  
- session_id: `02b9e9c9-3806-45ca-9b24-2ecc35a81efd`
```

Transcript

1. user (2026-06-10T16:28:02.244Z)

You are an adversarial verifier. A code reviewer audited the repo at C:/Users/avidu/Projects/muneem and reported this finding:

TITLE: Cr/Dr suffix sign-stripped in parse_amount ? an overdraft (Dr-balance) statement reconciles with every debit/credit INVERTED and persists as verified
SEVERITY CLAIMED: high
FILE: src/muneem/parsers/validation.py (line 32, 45)
DESCRIPTION: parse_amount strips a trailing 'Cr'/'Dr' and returns the magnitude with no sign: '1,500.00 Dr' -> +1500.0. For an account whose balance is negative throughout (OD facility / EMI overdue ? Dr-suffixed balances are how Indian banks print these), the magnitude sequence of a negative-balance walk is exactly a sign-flipped positive walk, so the L2 mapping search (engine.py:285-293) finds that assigning the Withdrawals column to 'credit' and Deposits to 'debit' makes the arithmetic work, and map_table returns it verified=True. Empirically confirmed on an AXIS_PROFILE header-less matrix (exactly what cluster_to_matrix emits, since txn_line_predicate drops header lines): true events 'withdraw 500, deposit 300, withdraw 800' on balances -1000->-1500->-1200->-2000 printed as '1,500.00 Dr' etc. came back `verified: True, layer: L2, map {'debit': 3, 'credit': 2}` ? every direction inverted, balances stored positive. The total-reconcile bookend does NOT save this even if finding #1 is fixed:
parse_statement_summary's regex also captures magnitudes only, and magnitudes are self-consistent under inversion. This is the worst-case class: wrong rows that still reconcile.
EVIDENCE QUOTED: validation.py:32: `\_CR\_DR\_SUFFIX = re.compile(r"(?i)\s\*(cr|dr)\s\*\$")`; :45: `text = \_CR\_DR\_SUFFIX.sub("", str(value).strip())...` (sign discarded). engine.py:43: `\_MONEY\_RE = re.compile(r"(?i)^\s\*[\d,]+\.\d{1,2}(?:\s\*(?:cr|dr))?\$")`. Empirical: map_table on Dr-suffixed balances returned `verified: True layer: L2 map: {'txn\_date': 0, 'narration': 1, 'debit': 3, 'credit': 2, 'balance': 4}` with 'UPI OUT' stored as credit=500.0.
SUGGESTED FIX: Make parse_amount sign-aware: a 'Dr' suffix on a balance yields a negative value (Cr positive), or return (magnitude, suffix) and have the engine reject candidate mappings that contradict the printed suffix. Additionally, have map_table detect AMBIGUITY: if more than one candidate mapping passes the oracle (the inverted and correct mappings both pass for sign-symmetric ledgers), refuse to auto-select and quarantine.

Your job is to try to REFUTE it. Your lens: correctness: trace the actual code path the finding describes and check the claimed behavior is what the code really does (read the file and its callers/tests).

Read the actual code in the repo (and tests/config where relevant). If after reading you cannot confirm the problem is real, set isReal=false. If it is real but over/under-stated, set adjustedSeverity accordingly. Be strict: plausible-sounding but unconfirmed = refuted.

SENSITIVE-DATA RULES (mandatory, highest priority):

- testdata/aubank-statements/, testdata/axis-statements/, testdata/hdfc-statements/, testdata/snapshots/, and testdata/credentials/ in the muneem repo contain the user's REAL financial documents and a real OAuth client file. Do NOT open/read the PDFs. You MAY check file existence and git-tracking status (git ls-files, git log --all -- <path>).
- If you must inspect testdata/snapshots/*.json or testdata/credentials/*.json to assess

sensitivity, read at most the JSON key structure and NEVER quote values (no account numbers, balances, person names, client secrets, transaction amounts) in your output.

- Never include sensitive values in your findings. Refer to them abstractly ("a real account number", "a client_secret value").
- Do not use WebSearch/WebFetch and do not send any repo content to any external service.

Structured output only.

2. user (2026-06-10T16:28:08.484Z)

```
1      """The **verifier** ? reconciliation as a layout-independent truth oracle.
2
3      Reconciliation proves a parse is **complete and correct without inspecting any sensitive
value**:.
4      if a transaction is missed or misread, the arithmetic stops adding up. That is what lets
us trust a
5      parse of a statement we never eyeball, and (later) what makes a probabilistic extractor
safe on
6      money ? *extraction proposes, reconciliation disposes* (see
7      [docs/parser-architecture.md](../docs/parser-architecture.md) §2, §5).
8
9      This module grew from a single running-balance walk into the verifier the architecture
calls for:
10
11     * **per-row walk** ? ``reconcile_running_balance``: ``balance[i] == balance[i-1] ? debit
+ credit``.
12     * **total reconcile** ? ``reconcile_total``: ``opening + ?credit ? ?debit == closing``
(work even
13     with no per-row balance, e.g. Axis savings).
14     * **bookend** ? ``verify``: parsed Dr/Cr counts + opening/closing vs the bank's printed
SUMMARY.
15     * **selection** ? ``select_reconciling``: among candidate concept mappings, pick the one
arithmetic
16     endorses (the seam the L2 mapping search plugs into).
17     * **confidence** ? ``Confidence`` + ``BookendReport.confidence``: a recorded verdict,
not an
18     assumption.
19
20     Reports carry counts / booleans / a verdict only ? **never an amount**.
21     """
22
23     from __future__ import annotations
24
25     import re
26     from collections.abc import Sequence
27     from dataclasses import dataclass
28     from enum import StrEnum
29
30     from .concepts import ConceptRow, StatementConcepts
31
32     _CR_DR_SUFFIX = re.compile(r"(?i)\s*(cr|dr)\s*$")
33
34
35     def parse_amount(value: object) -> float | None:
36         """Parse a statement money cell to a float.
37
38         Returns ``None`` only for a genuinely absent cell (None / empty / ``"-``" /
non-numeric); ``"0.00"`` parses to ``0.0``. This deliberately distinguishes
39         *absent* from *zero* ? unlike ``base.safe_amount``, which maps ``"0.00"`` to
40         ``None`` and would corrupt a real zero balance and the reconciliation math.
41         """
42
43         if value is None:
44             return None
45         text = _CR_DR_SUFFIX.sub("", str(value).strip()).replace(",", "").strip()
46         if text in ("", "-"):
47             return None
```

```

48         try:
49             return float(text)
50         except ValueError:
51             return None
52
53
54 @dataclass
55 class ReconcileReport:
56     """Result of a running-balance walk. Counts only, never an amount."""
57
58     n: int # rows considered
59     compared: int # consecutive (balance-bearing) pairs actually checked
60     breaks: int
61     first_break: int | None = None
62
63     @property
64     def ok(self) -> bool:
65         return self.compared > 0 and self.breaks == 0
66
67
68 def reconcile_running_balance(
69     entries: Sequence[tuple[float | None, float | None, float | None]],
70     tol: float = 0.01,
71 ) -> ReconcileReport:
72     """Verify ``balance[i] == balance[i-1] - debit[i] + credit[i]`` across rows.
73
74     ``entries`` is an ordered list of ``(debit, credit, balance)``; ``debit`` and
75     ``credit`` may be ``None`` (treated as 0.0). A row whose ``balance`` is ``None``
76     is skipped (can't anchor on it). Because the walk spans the whole statement, a
77     clean result also rules out any row missed *between* two balances ? which is the
78     failure mode silent ``extract_tables()`` row-drops would otherwise hide.
79     """
80     breaks = 0
81     first_break: int | None = None
82     compared = 0
83     prev: float | None = None
84     for i, (debit, credit, balance) in enumerate(entries):
85         if balance is None:
86             continue
87         if prev is not None:
88             expected = prev - (debit or 0.0) + (credit or 0.0)
89             compared += 1
90             if abs(expected - balance) > tol:
91                 breaks += 1
92                 if first_break is None:
93                     first_break = i
94         prev = balance
95     return ReconcileReport(
96         n=len(entries), compared=compared, breaks=breaks, first_break=first_break
97     )
98
99
100 def _row_entries(
101     rows: Sequence[ConceptRow],
102 ) -> list[tuple[float | None, float | None, float | None]]:
103     """Project concept rows onto the ``(debit, credit, balance)`` tuples the walk
104     consumes."""
105     return [(r.debit, r.credit, r.balance) for r in rows]
106
107
108 def reconcile_total(
109     rows: Sequence[ConceptRow], opening: float, closing: float, tol: float = 0.01
110 ) -> bool:
111     """Verify the layout-independent identity ``opening + ?credit ? ?debit == closing``.

```

```

112         This is the oracle for statements with **no reliable per-row balance** (e.g. Axis
savings):
113         it checks the whole statement against its own opening/closing bookends without
needing each
114         row's running balance. ``opening`` and ``closing`` come from the bank's
SUMMARY/header
115         (``StatementConcepts``); the caller must have them.
116         """
117         delta = sum(r.credit or 0.0 for r in rows) - sum(r.debit or 0.0 for r in rows)
118         return abs((opening + delta) - closing) <= tol
119
120
121     class Confidence(StrEnum):
122         """Recorded trust verdict for a parsed statement (mirrors ``documents.status``
values).
123
124         ``RECONCILED`` ? the parse passed every available check, so it is trusted.
``UNRECONCILED`` ?
125         a check failed (or none could run); the parse is suspect and is quarantined rather
than trusted.
126         """
127
128         RECONCILED = "reconciled"
129         UNRECONCILED = "unreconciled"
130
131
132     @dataclass
133     class BookendReport:
134         """A parse cross-checked against the bank's own SUMMARY. Booleans + counts only.
135
136         ``None`` for a check means the figure wasn't available to verify (not a failure).
``ok``
137         requires the per-row balance walk *plus* every available check to pass.
138         """
139
140         n: int
141         reconciles: bool
142         debit_count_ok: bool | None = None
143         credit_count_ok: bool | None = None
144         opening_ok: bool | None = None
145         closing_ok: bool | None = None
146
147         @property
148         def ok(self) -> bool:
149             checks = [self.debit_count_ok, self.credit_count_ok, self.opening_ok,
self.closing_ok]
150             return self.n > 0 and self.reconciles and all(c is True for c in checks)
151
152         @property
153         def confidence(self) -> Confidence:
154             return Confidence.RECONCILED if self.ok else Confidence.UNRECONCILED
155
156
157     def verify(
158         rows: Sequence[ConceptRow], concepts: StatementConcepts, tol: float = 0.01
159     ) -> BookendReport:
160         """Bookend a parse against the bank's printed SUMMARY: reconciliation + the printed
figures.
161
162         The per-row walk catches any mid-statement error/omission; the Dr/Cr counts catch
any
163         extra/missing row; the opening & closing balances catch a missing first/last row.
Each printed
164         figure is only checked when present (``None`` ? not checked). The opening balance is
checked by

```

```

165         implication (``balance[0] ? credit[0] + debit[0]`` should equal the printed
opening).
166         """
167         reconciles = reconcile_running_balance(_row_entries(rows), tol).ok
168         debit_count_ok = credit_count_ok = opening_ok = closing_ok = None
169         if concepts.debit_count is not None:
170             debit_count_ok = sum(1 for r in rows if (r.debit or 0.0) > 0) ==
concepts.debit_count
171         if concepts.credit_count is not None:
172             credit_count_ok = sum(1 for r in rows if (r.credit or 0.0) > 0) ==
concepts.credit_count
173         if rows and rows[-1].balance is not None and concepts.closing_balance is not None:
174             closing_ok = abs(rows[-1].balance - concepts.closing_balance) <= tol
175         if rows and rows[0].balance is not None and concepts.opening_balance is not None:
176             implied_opening = rows[0].balance - (rows[0].credit or 0.0) + (rows[0].debit or
0.0)
177             opening_ok = abs(implied_opening - concepts.opening_balance) <= tol
178         return BookendReport(
179             n=len(rows),
180             reconciles=reconciles,
181             debit_count_ok=debit_count_ok,
182             credit_count_ok=credit_count_ok,
183             opening_ok=opening_ok,
184             closing_ok=closing_ok,
185         )
186
187
188     def select_reconciling(candidates: Sequence[Sequence[ConceptRow]], tol: float = 0.01) ->
int | None:
189         """Among candidate concept-row mappings, return the index of the first that
reconciles.
190
191         This is the *selection* half of "extraction proposes, reconciliation disposes": when
the
192         extraction ladder can't be sure which column?concept assignment is right, it
proposes several
193         and lets the per-row walk pick the correct one. Returns ``None`` if none reconcile.
194
195         The candidate *generation* (the L2 mapping search) is Step 2 of the architecture
plan; this is
196         the verifier-side seam it plugs into. Selection here uses the per-row walk, so
candidates need a
197         running balance ? balance-less statements select via :func:`reconcile_total` at the
call site.
198         """
199         for i, rows in enumerate(candidates):
200             if reconcile_running_balance(_row_entries(rows), tol).ok:
201                 return i
202         return None
203
204
205     __all__ = [
206         "parse_amount",
207         "ReconcileReport",
208         "reconcile_running_balance",
209         "reconcile_total",
210         "Confidence",
211         "BookendReport",
212         "verify",
213         "select_reconciling",
214     ]
215

```

```
src\muneem\parsers\__init__.py
src\muneem\parsers\concepts.py
src\muneem\parsers\validation.py
src\muneem\parsers\clustering.py
src\muneem\parsers\profiles.py
src\muneem\parsers\savings.py
src\muneem\parsers\engine.py
src\muneem\parsers\base.py
src\muneem\parsers\au.py
src\muneem\parsers\axis.py
src\muneem\parsers\hdfc.py
```

4. user (2026-06-10T16:28:15.506Z)

```
1      """The generic concept-mapping engine ? columns in, concepts out, arithmetic decides.
2
3      This is the *generic-first* half of the concept-core architecture (see
4      [docs/parser-architecture.md](../docs/parser-architecture.md) §4, §6).
5      Bank-agnostic code that takes a **cell-matrix** (the shape `pdfplumber.extract_tables()`
returns,
6      or what a coordinate-clustering extractor emits) and maps its columns onto the canonical
concepts
7      via a cheap-first ladder, with a profile supplying priors:
8
9      * **L0 ? header lexicon.** Match header-row labels against the profile's concept?label
aliases.
10     * **profile-order.** If there is no header, fall back to the profile's known column
order (a prior).
11     * **L1/L2 ? content + search.** Classify columns by content (dates, money), then
enumerate the
12     plausible amount-column ? {debit, credit, balance} assignments and **keep the one the
13     reconciliation oracle endorses** ? the self-identifying balance column, found by
*which mapping
14     makes the math work* rather than by its label.
15
16     The engine is **family-agnostic**: it is parameterized by a :class:`FamilySpec` (the
target concept
17     names, which are date/amount/balance, and a typed-row builder) and an injected `oracle`
18     (`rows -> bool`). Only the *bank* family ships today; an instrument family
(equity/MF/ETF/bond ? see
19     the project scope) would reuse this control flow with its own concepts + oracle. Nothing
here logs a
20     value; provenance is the winning layer + column map.
21     """
22
23     from __future__ import annotations
24
25     import itertools
26     import re
27     from collections.abc import Callable, Sequence
28     from dataclasses import dataclass, field
29
30     from .concepts import ConceptRow
31     from .validation import parse_amount
32
33     Cell = object # a table cell: str | None (we coerce defensively)
34     Row = Sequence[Cell]
35
36     # Generic Indian-statement date shapes; a profile can pin its own to avoid false
positives.
37     _GENERIC_DATE_PATTERNS: tuple[re.Pattern, ...] = (
38         re.compile(r"^\d{2}[/-]\d{2}[/-]\d{4}$"), # dd/mm/yyyy or dd-mm-yyyy
39         re.compile(r"^\d{2}[- ][A-Za-z]{3}[- ][\d{2,4}$"), # dd-Mon-yyyy / dd Mon yy
40     )
41     # "Money" requires a decimal point so bare integers (e.g. a long reference number) are
```

```

NOT mistaken
42     # for an amount column during content classification.
43     _MONEY_RE = re.compile(r"(?i)^-?[\d,]+\.\d{1,2}(?:\s*(?:cr|dr))?$")
44     _BALANCE_MARKERS = ("opening balance", "closing balance", "balance b/f", "balance c/f")
45
46     # The generic bank lexicon. Profiles extend/override per issuer; keys are concept names,
values are
47     # lower-cased header labels matched exactly (not by substring) during L0.
48     _BANK_ALIASES: dict[str, tuple[str, ...]] = {
49         "txn_date": ("date", "txn date", "tran date", "transaction date", "posting date"),
50         "value_date": ("value date", "val date", "value dt"),
51         "narration": (
52             "narration",
53             "particulars",
54             "description",
55             "transaction details",
56             "details",
57             "remarks",
58             "transaction",
59             "transaction remarks",
60         ),
61         "reference": (
62             "ref",
63             "reference",
64             "ref no",
65             "reference no",
66             "cheque no",
67             "chq no",
68             "instrument no",
69         ),
70         "debit": ("debit", "withdrawal", "withdrawals", "dr", "paid out", "debit amount"),
71         "credit": ("credit", "deposit", "deposits", "cr", "paid in", "credit amount"),
72         "balance": (
73             "balance",
74             "closing balance",
75             "running balance",
76             "closing bal",
77             "available balance",
78         ),
79     }
80
81
82     def _text(cell: Cell) -> str:
83         return "" if cell is None else str(cell).strip()
84
85
86     def _norm(label: str) -> str:
87         return re.sub(r"\s+", " ", label.strip().lower())
88
89
90     def _matches_date(text: str, patterns: Sequence[re.Pattern]) -> bool:
91         return any(p.match(text) for p in patterns)
92
93
94     def _looks_like_money(text: str) -> bool:
95         return bool(_MONEY_RE.match(text))
96
97
98     @dataclass
99     class FamilySpec:
100         """A concept *family*: the target vocabulary + how to build typed rows from a column
mapping.
101
102         The engine's L0/L1/L2 logic is written against this spec rather than against bank
concepts, so a

```

```

103     future instrument family (with its own concepts + reconciliation oracle) reuses the
same engine.
104     ``build`` turns per-row ``{concept: raw_cell}`` dicts into typed rows the oracle
understands.
105     """
106
107     name: str
108     concepts: tuple[str, ...]
109     date_concepts: tuple[str, ...] # in order: the first is the primary (row-gating)
date
110     amount_concepts: tuple[str, ...] # money columns; the non-balance ones are the
"flows"
111     balance_concept: str | None # the running-balance concept, if the family has one
112     text_concepts: tuple[str, ...] # free-text columns; the first gets the widest text
column
113     default_aliases: dict[str, tuple[str, ...]]
114     build: Callable[[list[dict[str, str]]], list]
115
116
117     def _build_bank_rows(mapped: list[dict[str, str]]) -> list[ConceptRow]:
118         """Turn mapped cells into :class:`ConceptRow`s (amounts parsed; absent stays
``None``)."""
119         rows: list[ConceptRow] = []
120         for m in mapped:
121             rows.append(
122                 ConceptRow(
123                     txn_date=_text(m.get("txn_date")),
124                     narration=_text(m.get("narration")),
125                     debit=parse_amount(m.get("debit")),
126                     credit=parse_amount(m.get("credit")),
127                     balance=parse_amount(m.get("balance")),
128                     value_date=_text(m.get("value_date")) or None,
129                     reference=_text(m.get("reference")) or None,
130                 )
131             )
132         return rows
133
134
135     BANK_FAMILY = FamilySpec(
136         name="bank",
137         concepts=("txn_date", "value_date", "narration", "reference", "debit", "credit",
"balance"),
138         date_concepts=("txn_date", "value_date"),
139         amount_concepts=("debit", "credit", "balance"),
140         balance_concept="balance",
141         text_concepts=("narration", "reference"),
142         default_aliases=_BANK_ALIASES,
143         build=_build_bank_rows,
144     )
145
146
147     @dataclass
148     class MappingResult:
149         """The outcome of mapping one table: typed rows + how we got there (provenance)."""
150
151         rows: list # typed rows from family.build (e.g. list[ConceptRow])
152         mapped: list[
153             dict[str, str]
154         ] # per-row {concept: raw cell}, incl. any profile-declared extra columns
155         column_map: dict[str, int] # the winning concept -> column-index assignment
156         layer: str # "L0" / "profile-order" / "L2" ? which rung produced it
157         verified: bool # did the oracle endorse it? (False if no oracle was supplied)
158
159
160     @dataclass

```



```

161     class Profile:
162         """Declarative per-issuer knowledge ? *data, not code* (architecture §6).
163
164         Generic detectors run first; the profile supplies priors: extra concept?label
aliases, the
165         accepted date shape, a positional fallback when there's no header, table anchors,
account-number
166         patterns, quirks, and (placeholder, wiring parked) the password rule.
167         """
168
169         name: str
170         aliases: dict[str, tuple[str, ...]] = field(default_factory=dict)
171         date_patterns: tuple[re.Pattern, ...] = ()
172         column_order: tuple[str, ...] | None = None
173         table_anchors: tuple[str, ...] = ()
174         account_patterns: tuple[re.Pattern, ...] = ()
175         quirks: frozenset[str] = frozenset()
176         password_rule: str | None = None # data only; sourcing is parked (DESIGN §8.2)
177
178         def first_account(self, text: str) -> str | None:
179             for rx in self.account_patterns:
180                 m = rx.search(text)
181                 if m:
182                     return m.group(1).strip()
183             return None
184
185
186         def _alias_index(profile: Profile, family: FamilySpec) -> dict[str, str]:
187             """Build ``normalized-label -> concept`` (profile entries override family
defaults)."""
188             index: dict[str, str] = {}
189             for concept, labels in family.default_aliases.items():
190                 for label in labels:
191                     index[_norm(label)] = concept
192             for concept, labels in profile.aliases.items():
193                 for label in labels:
194                     index[_norm(label)] = concept
195             return index
196
197
198         def _date_patterns(profile: Profile) -> tuple[re.Pattern, ...]:
199             return profile.date_patterns or _GENERIC_DATE_PATTERNS
200
201
202         def _detect_header(
203             rows: list[list[Cell]], alias_index: dict[str, str]
204         ) -> tuple[int | None, dict[str, int]]:
205             """Find the header row (most exact alias hits, >=2) and its concept->column map."""
206             best_idx: int | None = None
207             best_map: dict[str, int] = {}
208             for i, row in enumerate(rows):
209                 cmap: dict[str, int] = {}
210                 for col, cell in enumerate(row):
211                     concept = alias_index.get(_norm(_text(cell)))
212                     if concept and concept not in cmap:
213                         cmap[concept] = col
214                 if len(cmap) > len(best_map):
215                     best_map, best_idx = cmap, i
216             if len(best_map) < 2:
217                 return None, {}
218             return best_idx, best_map
219
220
221         def _is_balance_marker(row: Row) -> bool:
222             return any(any(mk in _norm(_text(c)) for mk in _BALANCE_MARKERS) for c in row)

```

```

223
224
225 def _is_transaction_row(row: Row, date_col: int | None, patterns: Sequence[re.Pattern])
-> bool:
226     if date_col is not None:
227         cell = row[date_col] if date_col < len(row) else None
228         return _matches_date(_text(cell), patterns)
229     return any(_matches_date(_text(c), patterns) for c in row)
230
231
232 def _classify_columns(
233     data: list[list[Cell]], ncols: int, patterns: Sequence[re.Pattern]
234 ) -> dict[int, str]:
235     """Tag each column date / amount / text / empty by majority content."""
236     kinds: dict[int, str] = {}
237     for c in range(ncols):
238         cells = [_text(r[c]) for r in data if c < len(r) and _text(r[c])]
239         if not cells:
240             kinds[c] = "empty"
241             continue
242         thresh = max(1, int(len(cells) * 0.6))
243         if sum(1 for x in cells if _matches_date(x, patterns)) >= thresh:
244             kinds[c] = "date"
245         elif sum(1 for x in cells if _looks_like_money(x)) >= thresh:
246             kinds[c] = "amount"
247         else:
248             kinds[c] = "text"
249     return kinds
250
251
252 def _base_map(
253     col_kinds: dict[int, str], data: list[list[Cell]], family: FamilySpec
254 ) -> dict[str, int]:
255     """The unambiguous part of a content (L1) mapping: date columns + the primary text
column.
256
257     Family-agnostic: date columns fill ``family.date_concepts`` in order; the widest
text column
258     goes to the family's first text concept (if any). Amount columns are left to the
search.
259
260     """
261     base: dict[str, int] = {}
262     date_cols = [c for c, k in sorted(col_kinds.items()) if k == "date"]
263     text_cols = [c for c, k in sorted(col_kinds.items()) if k == "text"]
264     for concept, col in zip(family.date_concepts, date_cols, strict=False):
265         base[concept] = col
266     if text_cols and family.text_concepts:
267         base[family.text_concepts[0]] = max(
268             text_cols, key=lambda c: sum(len(_text(r[c])) for r in data if c < len(r))
269         )
270     return base
271
272 def _search_candidates(
273     base: dict[str, int], col_kinds: dict[int, str], family: FamilySpec
274 ) -> list[tuple[str, dict[str, int]]]:
275     """Enumerate plausible amount-column ? concept assignments (the L2 search space).
276
277     Family-agnostic: the "flows" are the family's amount concepts minus its balance
concept (for
278     banks: debit, credit). We try each amount column as the balance and assign the rest
to the
279     flows, then also the no-per-row-balance case (total-reconcile families, e.g. Axis
Oct).
280     """

```

```

281     amount_cols = [c for c, k in sorted(col_kinds.items()) if k == "amount"]
282     flows = [c for c in family.amount_concepts if c != family.balance_concept]
283     bal = family.balance_concept
284     cands: list[dict[str, int]] = []
285     if bal and len(amount_cols) >= len(flows) + 1:
286         for bcol in amount_cols:
287             rest = [c for c in amount_cols if c != bcol]
288             for assign in itertools.permutations(rest, len(flows)):
289                 cands.append(**base, **dict(zip(flows, assign, strict=True)), bal:
bcol})
290     if flows and len(amount_cols) >= len(flows):
291         for assign in itertools.permutations(amount_cols, len(flows)):
292             cands.append(**base, **dict(zip(flows, assign, strict=True)))
293     return [("L2", c) for c in cands[:24]]
294
295
296     def _apply_map(data: list[list[Cell]], col_map: dict[str, int]) -> list[dict[str, str]]:
297         out: list[dict[str, str]] = []
298         for row in data:
299             out.append({c: (_text(row[i]) if i < len(row) else "") for c, i in
col_map.items()})
300         return out
301
302
303     def map_table(
304         table: Sequence[Row] | None, # Sequence (covariant) so list[list[str | None]]
callers fit
305         profile: Profile,
306         family: FamilySpec = BANK_FAMILY,
307         *,
308         oracle: Callable[[list], bool] | None = None,
309     ) -> MappingResult | None:
310         """Map one extracted table onto ``family``'s concepts; ``oracle`` selects among
candidates.
311
312         Resolution order: header lexicon (L0) ? profile column order ? content search (L2).
With an
313         ``oracle`` the first candidate it endorses wins (``verified=True``); without one,
the first
314         structurally-complete candidate is returned unverified (the caller reconciles
separately).
315         Returns ``None`` if no transaction rows are found, or the best unverified candidate
if an oracle
316         was supplied but nothing reconciled.
317         """
318         if not table:
319             return None
320         rows_in = [list(r) for r in table if r is not None]
321         if not rows_in:
322             return None
323         alias_index = _alias_index(profile, family)
324         patterns = _date_patterns(profile)
325
326         header_idx, col_map_hdr = _detect_header(rows_in, alias_index)
327         start = header_idx + 1 if header_idx is not None else 0
328         date_col_hint = col_map_hdr.get("txn_date")
329         if date_col_hint is None and profile.column_order and "txn_date" in
profile.column_order:
330             date_col_hint = profile.column_order.index("txn_date")
331
332         data = [
333             r
334             for r in rows_in[start:]
335             if not _is_balance_marker(r) and _is_transaction_row(r, date_col_hint, patterns)
336         ]

```

```

337         if not data:
338             return None
339         ncols = max(len(r) for r in data)
340         col_kinds = _classify_columns(data, ncols, patterns)
341         base = _base_map(col_kinds, data, family)
342         flows = [c for c in family.amount_concepts if c != family.balance_concept]
343
344         candidates: list[tuple[str, dict[str, int]]] = []
345         if flows and all(c in col_map_hdr for c in flows): # L0: header pinned the flow
concepts
346             candidates.append(("L0", {**base, **col_map_hdr}))
347         elif col_map_hdr: # header present but didn't pin the flows: keep its hints, search
the rest
348             candidates.append(("L0", {**base, **col_map_hdr}))
349         if profile.column_order: # positional prior (no/partial header)
350             candidates.append(("profile-order", {c: i for i, c in
enumerate(profile.column_order)}))
351         candidates += _search_candidates(base, col_kinds, family)
352
353         primary_date = family.date_concepts[0] if family.date_concepts else None
354         best: MappingResult | None = None
355         for layer, cmap in candidates:
356             if primary_date and primary_date not in cmap:
357                 continue
358             if not (set(family.amount_concepts) & cmap.keys()):
359                 continue
360             mapped = _apply_map(data, cmap)
361             typed = family.build(mapped)
362             if oracle is None:
363                 return MappingResult(typed, mapped, cmap, layer, verified=False)
364             if oracle(typed):
365                 return MappingResult(typed, mapped, cmap, layer, verified=True)
366             if best is None:
367                 best = MappingResult(typed, mapped, cmap, layer, verified=False)
368         return best
369
370
371     __all__ = ["Profile", "FamilySpec", "BANK_FAMILY", "MappingResult", "map_table"]
372

```

5. user (2026-06-10T16:28:40.894Z)

```

1     """Per-issuer profiles ? bank knowledge expressed as **data, not code** (architecture
$6).
2
3     A profile carries the priors the generic engine needs to disambiguate one issuer: extra
4     concept?label aliases, the accepted date shape, a positional fallback when there's no
header,
5     table anchors, account-number patterns, quirks, and (parked) the password rule. Adding
an issuer is
6     ideally *adding a profile*, not writing a parser.
7
8     `HDFC_PROFILE` is data consumed by HDFC's parser, whose *extraction* stays bespoke
coordinate-
9     clustering ? the documented exception where `extract_tables()` fails on a semi-ruled
table (see the
10    `hdfc-parser-tech-decision` memory). The Axis and AU Bank profiles land with their
engine-driven
11    parsers (coordinate clustering ? generic mapping ? total reconcile).
12    """
13
14    from __future__ import annotations
15
16    import re
17

```

```

18     from .engine import Profile
19
20     AXIS_PROFILE = Profile(
21         name="axis",
22         aliases={
23             # Real Axis savings header: Tran/Txn Date | Transaction | Withdrawals | Deposits
24             | Balance |
25             # Other Information. "Transaction" is the narration column (not the date).
26             "narration": ("transaction", "particulars"),
27             "debit": ("withdrawals", "withdrawal", "debit"),
28             "credit": ("deposits", "deposit", "credit"),
29             "balance": ("balance",),
30         },
31         date_patterns=(re.compile(r"^\d{2}[-/]\d{2}[-/]\d{4}$"),), # dd-mm-yyyy or
32         dd/mm/yyyy
33         # Layout / column-count varies between vintages, so there is NO reliable
34         column_order ? the
35         # generic content search + total reconcile do the work. Broadened account pattern:
36         the old
37         # `Account No:` regex missed real statements.
38         account_patterns=(
39             re.compile(r"Account\s*(?:No|Number)\s*[:\.-]?s*(\d{6,})", re.IGNORECASE),
40             re.compile(r"A/c\s*No\.\.?s*[:\s]*(\d{6,})", re.IGNORECASE),
41         ),
42         quirks=frozenset({"layout-varies", "other-information-column", "balance-optional"}),
43     )
44
45     AUBANK_PROFILE = Profile(
46         name="aubank",
47         aliases={
48             "narration": (
49                 "transaction",
50                 "particulars",
51                 "description",
52                 "narration",
53                 "transaction details",
54             ),
55             "debit": ("withdrawal", "withdrawals", "debit", "withdrawal amount", "dr"),
56             "credit": ("deposit", "deposits", "credit", "deposit amount", "cr"),
57             "balance": ("balance", "closing balance", "running balance"),
58         },
59         # AU vintages use dd-mm-yyyy / dd/mm/yyyy or dd-Mon-yyyy; verified on the real
60         statements.
61         date_patterns=(
62             re.compile(r"^\d{2}[-/]\d{2}[-/]\d{4}$"),
63             re.compile(r"^\d{2}-[A-Za-z]{3}-\d{4}$"),
64         ),
65         account_patterns=(
66             re.compile(r"Account\s*(?:No|Number)\s*[:\.-]?s*(\d{6,})", re.IGNORECASE),
67             re.compile(r"A/c\s*No\.\.?s*[:\s]*(\d{6,})", re.IGNORECASE),
68         ),
69         quirks=frozenset({"engine-driven"}),
70     )
71
72     HDFC_PROFILE = Profile(
73         name="hdfc",
74         aliases={
75             "debit": ("withdrawal", "withdrawals"),
76             "credit": ("deposit", "deposits"),
77             "balance": ("closing balance",),
78         },
79         date_patterns=(re.compile(r"^\d{2}/\d{2}/\d{4}$"),), # dd/mm/yyyy
80         table_anchors=("Narration", "Closing Balance"),
81         account_patterns=(re.compile(r"Account\s*Number\s*[:.]?s*(\d{6,})",
82 re.IGNORECASE),),

```

```

77         # HDFC's transaction table is semi-ruled; extract_tables() is unreliable on it, so
HDFC extracts
78         # by word-position clustering rather than through the generic cell-matrix engine.
79         quirks=frozenset({"semi-ruled-table", "coordinate-clustering"}),
80     )
81
82     __all__ = ["AXIS_PROFILE", "AUBANK_PROFILE", "HDFC_PROFILE"]
83

```

6. user (2026-06-10T16:28:41.138Z)

```

1     """Coordinate clustering ? turn positioned words into a clean cell-matrix.
2
3     When ``pdfplumber.extract_tables()`` mis-groups a semi-ruled or whitespace-aligned
statement (the
4     Axis case: a date merged into the narration cell, column counts drifting between
vintages), we read
5     **word positions** directly. Words are grouped into visual rows by their ``top``; column
boundaries
6     are inferred from the vertical whitespace **shared across the transaction rows**; and
each word is
7     dropped into the column its x-centre falls in. Because columns come from geometry, a
leading date
8     sits in its *own* column instead of being glued to the narration.
9
10    Two robustness lessons baked in:
11
12    * **Infer columns from transaction-like rows only.** Headers, address blocks, and
summaries span the
13    page and would erase the inter-column whitespace. A caller-supplied ``keep_line``
predicate (e.g.
14    "has a date *and* a money token") restricts both column inference and the emitted
matrix to the
15    transaction band.
16    * **Use a coverage histogram, not interval-merging.** A column boundary is an x-channel
empty
17    in *most* rows; a single long narration can't collapse two columns.
18
19    Bank-agnostic extraction (no issuer constants, no x-positions hard-coded); the engine's
oracle still
20    decides whether the result is trustworthy. Pure geometry in, strings out ? no value is
logged.
21    """
22
23    from __future__ import annotations
24
25    from collections.abc import Callable, Sequence
26
27    Word = dict # a pdfplumber word: {"text", "x0", "x1", "top", "bottom", ...}
28
29
30    def group_lines(words: Sequence[Word], tol: float = 2.5) -> list[list[Word]]:
31        """Group words into visual lines by ``top`` (tolerant of sub-pixel jitter), left-to-
right."""
32        lines: list[list[Word]] = []
33        anchor_top: float | None = None
34        for w in sorted(words, key=lambda w: (w["top"], w["x0"])):
35            if lines and anchor_top is not None and abs(w["top"] - anchor_top) <= tol:
36                lines[-1].append(w)
37            else:
38                lines.append([w])
39                anchor_top = w["top"]
40        return [sorted(line, key=lambda w: w["x0"]) for line in lines]
41
42

```

```

43     def infer_columns(
44         lines: Sequence[Sequence[Word]], bin_w: float = 1.0, min_frac: float = 0.3
45     ) -> list[tuple[float, float]]:
46         """Infer column x-extents from a coverage histogram over ``lines``.
47
48         An x-bin is "filled" if at least ``min_frac`` of the lines have a word covering it;
columns are
49         the runs of filled bins, and the empty runs between them are the inter-column
whitespace. Robust
50         to a few full-width lines: it thresholds on the *fraction* of rows, not on any
single one.
51         """
52         words = [w for line in lines for w in line]
53         if not words:
54             return []
55         x0 = min(float(w["x0"]) for w in words)
56         x1 = max(float(w["x1"]) for w in words)
57         nbins = max(1, int((x1 - x0) / bin_w) + 1)
58         cover = [0] * nbins
59         for line in lines:
60             marked: set[int] = set()
61             for w in line:
62                 a = int((float(w["x0"]) - x0) / bin_w)
63                 b = int((float(w["x1"]) - x0) / bin_w)
64                 marked.update(range(max(0, a), min(nbins - 1, b) + 1))
65             for k in marked:
66                 cover[k] += 1
67         thresh = max(1.0, min_frac * len(lines))
68         filled = [c >= thresh for c in cover]
69         cols: list[tuple[float, float]] = []
70         start: int | None = None
71         for i, f in enumerate(filled):
72             if f and start is None:
73                 start = i
74             elif not f and start is not None:
75                 cols.append((x0 + start * bin_w, x0 + i * bin_w))
76                 start = None
77         if start is not None:
78             cols.append((x0 + start * bin_w, x0 + nbins * bin_w))
79         return cols
80
81
82     def _column_of(center: float, columns: list[tuple[float, float]]) -> int:
83         for i, (a, b) in enumerate(columns):
84             if a <= center <= b:
85                 return i
86         return min(
87             range(len(columns)),
88             key=lambda i: min(abs(center - columns[i][0]), abs(center - columns[i][1])),
89         )
90
91
92     def line_to_cells(line: Sequence[Word], columns: list[tuple[float, float]]) ->
list[str]:
93         """Place each word into its column; join words that share a column (in x order)."""
94         cells = [""] for _ in columns
95         for w in line:
96             idx = _column_of((float(w["x0"]) + float(w["x1"])) / 2, columns)
97             cells[idx] = (cells[idx] + " " + str(w["text"])).strip()
98         return cells
99
100
101     def cluster_to_matrix(
102         words_by_page: Sequence[Sequence[Word]],
103         *,

```

```

104     line_tol: float = 2.5,
105     min_frac: float = 0.3,
106     keep_line: Callable[[list[str]], bool] | None = None,
107 ) -> list[list[str]]:
108     """Cluster words across pages into one cell-matrix with consistent columns.
109
110     ``keep_line`` (texts of a line ? bool) restricts both column inference and the
output to the
111     transaction band ? without it, full-width headers/footers swamp the column
structure. Columns
112     are inferred once across all kept lines (a statement keeps its layout); rows are
emitted in
113     page-then-top order.
114     """
115     lines: list[list[Word]] = []
116     for page in words_by_page:
117         lines.extend(group_lines(page, line_tol))
118     if keep_line is not None:
119         lines = [ln for ln in lines if keep_line([str(w["text"]) for w in ln])]
120     if not lines:
121         return []
122     columns = infer_columns(lines, min_frac=min_frac)
123     if not columns:
124         return []
125     return [line_to_cells(line, columns) for line in lines]
126
127
128 __all__ = ["group_lines", "infer_columns", "line_to_cells", "cluster_to_matrix"]
129

```

7. user (2026-06-10T16:28:50.200Z)

```

1     """Shared engine-driven savings-statement parsing (Axis, AU Bank, and future issuers).
2
3     Every issuer whose statement is a transaction table follows the same recipe, so it lives
here once
4     and each bank supplies only a :class:`~muneem.parsers.engine.Profile`:
5
6     1. Extraction proposes candidate cell-matrices ? one per ``extract_tables()`` table,
plus a
7     coordinate-clustered whole-statement matrix (for layouts ``extract_tables`` mis-
segments).
8     2. The generic engine maps each candidate's columns onto the canonical concepts.
9     3. Reconciliation disposes: a parse is trusted only if its per-row balance walks, or
? when a
10     vintage has no per-row balance ? the totals reconcile to the printed
``opening``/``closing``.
11
12     A statement splits its ledger across several extracted tables (page breaks), so we map
each table
13     on its own, keep the rows the oracle endorses, and re-check the combined set (which
catches a
14     cross-table balance break, i.e. a missing transaction). The result is oracle-gated:
a layout we
15     can't yet read returns ``verified=False`` and the caller persists nothing ? never
misabeled rows.
16
17     No statement content is logged; only counts/booleans travel out.
18     """
19
20     from __future__ import annotations
21
22     import re
23     from dataclasses import dataclass, field
24     from pathlib import Path

```



```

25
26 import pdfplumber
27
28 from .clustering import cluster_to_matrix
29 from .concepts import ConceptRow, StatementConcepts
30 from .engine import Profile, map_table
31 from .validation import parse_amount, reconcile_running_balance, reconcile_total
32
33 _MONEY_TOKEN = re.compile(r"^~?[\d,]+\.\d{2}(?:cr|dr)?$", re.IGNORECASE)
34 _OPENING_RE = re.compile(r"opening\s+balance[^\d-]*(-?[\d,]+\.\d{2})", re.IGNORECASE)
35 _CLOSING_RE = re.compile(r"closing\s+balance[^\d-]*(-?[\d,]+\.\d{2})", re.IGNORECASE)
36
37
38 @dataclass
39 class SavingsParse:
40     """Outcome of a savings parse: concept rows + whether reconciliation endorsed
41     them."""
42     account_number: str
43     rows: list[ConceptRow] = field(default_factory=list)
44     verified: bool = False
45
46
47 def parse_statement_summary(text: str) -> StatementConcepts:
48     """Read opening/closing balance from statement text (for the total-reconcile
49     oracle).
50
51     Best-effort regex; ``None`` when not found ? the oracle then relies on the per-row
52     walk.
53
54     """
55     opening = _OPENING_RE.search(text)
56     closing = _CLOSING_RE.search(text)
57     return StatementConcepts(
58         opening_balance=parse_amount(opening.group(1)) if opening else None,
59         closing_balance=parse_amount(closing.group(1)) if closing else None,
60     )
61
62
63 def txn_line_predicate(profile: Profile):
64     """A line is transaction-like if it carries one of the profile's dates *and* a money
65     token.
66
67     Used to restrict coordinate clustering to the transaction band (drops headers /
68     summaries /
69     address blocks that would otherwise wreck column inference).
70
71     """
72     patterns = profile.date_patterns
73
74     def keep(texts: list[str]) -> bool:
75         has_date = any(any(p.match(t) for p in patterns) for t in texts)
76         return has_date and any(_MONEY_TOKEN.match(t) for t in texts)
77
78     return keep
79
80
81 def reconcile_oracle(opening: float | None, closing: float | None):
82     """Trust a parse if the per-row balance walks, else if the totals reconcile to the
83     bookends."""
84
85     def check(rows: list[ConceptRow]) -> bool:
86         if reconcile_running_balance([(r.debit, r.credit, r.balance) for r in rows]).ok:
87             return True
88         if rows and opening is not None and closing is not None:
89             return reconcile_total(rows, opening, closing)
90         return False

```

```

84
85     return check
86
87
88     def _reconstruct_by_balance_delta(
89         matrix: list[list[str]], profile: Profile, opening: float | None, tol: float = 0.01
90     ) -> list[ConceptRow]:
91         """Recover debit/credit from the running balance when columns won't split cleanly.
92
93         Needs only three things per row ? a date, the transaction amount, and the running
94         balance ? so
95         it tolerates messy column inference (a sliver column, an interleaved value-date
96         column, or a
97         single merged withdrawal/deposit column). Direction is the sign of the balance
98         change; the
99         caller's reconciliation then **cross-checks** that the amount equals |?balance| on
100        every row, so
101        a misread or a dropped row still fails (it is *not* circular). Needs the printed
102        opening balance
103        to fix the first row's direction; returns ``[]`` if it can't form a clean series.
104        """
105        patterns = profile.date_patterns or (re.compile(r"^\d{2}[-/]\d{2}[-/]\d{4}$"),)
106        rows = [(str(c).strip() if c else "") for c in r] for r in (matrix or []) if r]
107        if not rows or opening is None:
108            return []
109        ncols = max(len(r) for r in rows)
110
111        def is_date(s: str) -> bool:
112            return any(p.match(s) for p in patterns)
113
114        def money(s: str) -> float | None:
115            return parse_amount(s) if _MONEY_TOKEN.match(s) else None
116
117        # date column = the leftmost of the (first few) columns with the most date cells
118        date_counts: dict[int, int] = {}
119        for r in rows:
120            for c in range(min(ncols, 3)):
121                if c < len(r) and is_date(r[c]):
122                    date_counts[c] = date_counts.get(c, 0) + 1
123        if not date_counts:
124            return []
125        top = max(date_counts.values())
126        date_col = min(c for c, n in date_counts.items() if n == top)
127
128        txn_rows = [r for r in rows if date_col < len(r) and is_date(r[date_col])]
129        if len(txn_rows) < 2:
130            return []
131
132        # A money column's money cells must outnumber its date cells ? so a sparsely-filled
133        deposit
134        # column still counts, but an interleaved value-date column (mostly dates) does not.
135        The
136        # running balance is the rightmost money column; the rest are the flow
137        (withdrawal/deposit).
138        money_cols = []
139        for c in range(ncols):
140            m = sum(1 for r in txn_rows if c < len(r) and money(r[c]) is not None)
141            d = sum(1 for r in txn_rows if c < len(r) and is_date(r[c]))
142            if m >= 1 and m > d:
143                money_cols.append(c)
144        if len(money_cols) < 2:
145            return []
146        balance_col = max(money_cols)
147        flow_cols = [c for c in money_cols if c != balance_col]

```

```

141     text_cols = [c for c in range(ncols) if c != date_col and c not in money_cols]
142     narr_col = (
143         max(text_cols, key=lambda c: sum(len(r[c]) for r in txn_rows if c < len(r)))
144         if text_cols
145         else None
146     )
147
148     out: list[ConceptRow] = []
149     prev = opening
150     for r in txn_rows:
151         bal = money(r[balance_col]) if balance_col < len(r) else None
152         if bal is None:
153             return [] # a running balance on every row is required for the delta
154         amt = next((money(r[c]) for c in flow_cols if c < len(r) and money(r[c]) is not
None), None)
155         debit = credit = None
156         if amt is not None:
157             if bal < prev - tol:
158                 debit = amt
159             elif bal > prev + tol:
160                 credit = amt
161             else:
162                 debit = amt # zero net with an amount present ? reconciliation will
flag it
163         out.append(
164             ConceptRow(
165                 txn_date=r[date_col],
166                 narration=(r[narr_col] if narr_col is not None and narr_col < len(r)
else ""),
167                 debit=debit,
168                 credit=credit,
169                 balance=bal,
170             )
171         )
172         prev = bal
173     return out
174
175
176     def parse_savings(pdf_path: Path, profile: Profile, password: str | None = None) ->
SavingsParse:
177         """Parse a savings statement into concept rows + a reconciliation verdict, per the
profile.
178
179         Per-table mapping first (each extracted table mapped on its own; oracle-endorsed
rows combined
180         and re-checked), then a coordinate-clustering fallback. Returns ``verified=False``
if neither
181         reconciles, so the caller can refuse to persist.
182         """
183         with pdfplumber.open(pdf_path, password=password) as pdf:
184             words_by_page = [
185                 page.extract_words(use_text_flow=False, keep_blank_chars=False) for page in
pdf.pages
186             ]
187             tables = [t for page in pdf.pages for t in (page.extract_tables() or [])]
188             head = "\n".join((pdf.pages[i].extract_text() or "") for i in range(min(3,
len(pdf.pages))))
189
190             account_number = profile.first_account(head) or "UNKNOWN"
191             summary = parse_statement_summary(head)
192             oracle = reconcile_oracle(summary.opening_balance, summary.closing_balance)
193
194             confirmed: list[ConceptRow] = []
195             for table in tables:
196                 res = map_table(table, profile, oracle=oracle)

```

```

197         if res and res.verified:
198             confirmed.extend(res.rows)
199     if confirmed and oracle(confirmed):
200         return SavingsParse(account_number, confirmed, True)
201
202     matrix = cluster_to_matrix(words_by_page, keep_line=txn_line_predicate(profile))
203     clustered = map_table(matrix, profile, oracle=oracle)
204     if clustered and clustered.verified:
205         return SavingsParse(account_number, clustered.rows, True)
206
207     # 3) balance-delta reconstruction ? recovers vintages whose columns don't split into
a clean
208     #     debit/credit (merged amount column, interleaved value-date, sliver columns):
derive the
209     #     direction from the running balance and let the oracle cross-check amount ==
|?balance|.
210     delta = _reconstruct_by_balance_delta(matrix, profile, summary.opening_balance)
211     if delta and oracle(delta):
212         return SavingsParse(account_number, delta, True)
213
214     fallback = confirmed or (clustered.rows if clustered else [])
215     return SavingsParse(account_number, fallback, False)
216
217
218     __all__ = [
219         "SavingsParse",
220         "parse_savings",
221         "parse_statement_summary",
222         "txn_line_predicate",
223         "reconcile_oracle",
224     ]
225

```

8. assistant (2026-06-10T16:29:25.747Z)

Now let me check the issuer parsers and how results persist, plus any tests covering Cr/Dr.

9. user (2026-06-10T16:29:26.744Z)

```

1     """AU Small Finance Bank savings / transaction-account statement parser.
2
3     Runs on the **shared engine-driven path** (:mod:`muneem.parsers.savings`): per-table +
4     coordinate-clustered candidates ? engine mapping ? reconciliation oracle, **oracle-
gated** so a
5     vintage it can't read stores nothing rather than mislabeled rows. The AU specifics are
entirely in
6     ``AUBANK_PROFILE`` ? this parser is mostly the DB wiring. No statement content is logged
here.
7     """
8
9     from __future__ import annotations
10
11     import sqlite3
12     from pathlib import Path
13
14     from muneem import __version__
15     from muneem import db as dbmod
16     from muneem.errors import ParseError
17     from muneem.redaction import redact_filename
18
19     from . import register_parser
20     from .base import StatementParser
21     from .profiles import AUBANK_PROFILE
22     from .savings import SavingsParse, parse_savings
23     from .validation import Confidence

```

```

24
25
26     def parse_au_savings(pdf_path: Path, password: str | None = None) -> SavingsParse:
27         """Parse an AU Bank savings statement via the shared engine-driven path."""
28         return parse_savings(pdf_path, AUBANK_PROFILE, password)
29
30
31     @register_parser
32     class AUParser(StatementParser):
33         def can_handle(self, text: str, sender: str | None = None, subject: str | None =
None) -> bool:
34             if sender and "aubank" in sender.lower():
35                 return True
36             low = (text or "").lower()
37             return "au small finance bank" in low or "aubank" in low
38
39         def parse_and_insert(
40             self, pdf_path: Path, conn: sqlite3.Connection, doc_id: int, password: str |
None = None
41         ) -> int:
42             """Persist a savings parse **only if it reconciles**; otherwise store
nothing."""
43             try:
44                 parsed = parse_au_savings(pdf_path, password)
45             except ParseError:
46                 raise
47             except Exception as exc:
48                 raise ParseError(f"failed to parse AU statement
{redact_filename(pdf_path)}") from exc
49
50             if not parsed.verified:
51                 dbmod.record_parse(
52                     conn,
53                     doc_id,
54                     confidence=Confidence.UNRECONCILED.value,
55                     parser="AUParser",
56                     parser_version=__version__,
57                 )
58             return 0
59
60             for r in parsed.rows:
61                 dbmod.insert_au_transaction(
62                     conn,
63                     document_id=doc_id,
64                     account_number=parsed.account_number,
65                     txn_date=r.txn_date,
66                     value_date=r.value_date or "",
67                     particulars=r.narration,
68                     debit=r.debit,
69                     credit=r.credit,
70                     balance=r.balance,
71                 )
72                 dbmod.record_parse(
73                     conn,
74                     doc_id,
75                     confidence=Confidence.RECONCILED.value,
76                     parser="AUParser",
77                     parser_version=__version__,
78                 )
79             return len(parsed.rows)
80

```

10. user (2026-06-10T16:29:27.701Z)

```

1         """Axis Bank statement parsers: savings/transaction account + credit card.

```

```

2
3     **Savings runs on the shared engine-driven path** (:mod:`muneem.parsers.savings`): per-
table +
4     coordinate-clustered candidates ? engine mapping ? reconciliation oracle, **oracle-
gated** so a
5     vintage it can't read stores nothing (`unreconciled`) rather than mislabeled rows.
Axis specifics
6     live entirely in `AXIS_PROFILE`; this replaced the old fixed-column `extract_tables`
mapping.
7
8     Credit-card statements carry no running balance, so `AxisCreditCardParser` can't
reconcile
9     (parse/persist split + count only). No statement content is logged here.
10    ""
11
12    from __future__ import annotations
13
14    import re
15    import sqlite3
16    from dataclasses import dataclass
17    from pathlib import Path
18
19    import pdfplumber
20
21    from muneem import __version__
22    from muneem import db as dbmod
23    from muneem.errors import ParseError
24    from muneem.redaction import redact_filename
25
26    from . import register_parser
27    from .base import StatementParser
28    from .profiles import AXIS_PROFILE
29    from .savings import SavingsParse, parse_savings, parse_statement_summary,
txn_line_predicate
30    from .validation import Confidence, parse_amount
31
32    _CC_DATE_RE = re.compile(r"\d{2}/\d{2}/\d{4}") # Axis credit card: dd/mm/yyyy
33    _CARD_RES = [re.compile(r"(\d{4}XXXX\d{4})"), re.compile(r"(\d{6}\*\d{4})")]
34
35
36    def _first_match(text: str, patterns: list[re.Pattern]) -> str | None:
37        for rx in patterns:
38            m = rx.search(text)
39            if m:
40                return m.group(1).strip()
41        return None
42
43
44    # --- Savings / transaction account (concept core)
-----
45
46
47    # Back-compat names; the savings logic now lives in muneem.parsers.savings (shared with
AU Bank).
48    AxisSavings = SavingsParse
49    parse_axis_summary = parse_statement_summary
50    _is_txn_line = txn_line_predicate(AXIS_PROFILE)
51
52
53    def parse_axis_savings(pdf_path: Path, password: str | None = None) -> SavingsParse:
54        """Parse an Axis savings statement via the shared engine-driven path
(savings.parse_savings)."""
55        return parse_savings(pdf_path, AXIS_PROFILE, password)
56
57

```

```

58     @register_parser
59     class AxisParser(StatementParser):
60         def can_handle(self, text: str, sender: str | None = None, subject: str | None =
None) -> bool:
61             if sender and "axisbank.com" in sender.lower():
62                 return True
63             t = text or ""
64             return "Axis Bank" in t and ("Statement of Account" in t or "Account Statement"
in t)
65
66         def parse_and_insert(
67             self, pdf_path: Path, conn: sqlite3.Connection, doc_id: int, password: str |
None = None
68         ) -> int:
69             """Persist a savings parse **only if it reconciles**; otherwise store nothing.
70
71             Sets ``documents.status`` to ``reconciled`` / ``unreconciled`` accordingly, so a
layout the
72             parser can't yet read surfaces loudly instead of saving mislabeled rows.
73             """
74             try:
75                 parsed = parse_axis_savings(pdf_path, password)
76             except ParseError:
77                 raise
78             except Exception as exc:
79                 # Filename only ? never statement text; original cause kept via `from`.
80                 name = redact_filename(pdf_path)
81                 raise ParseError(f"failed to parse Axis statement {name}") from exc
82
83             if not parsed.verified:
84                 dbmod.record_parse(
85                     conn,
86                     doc_id,
87                     confidence=Confidence.UNRECONCILED.value,
88                     parser="AxisParser",
89                     parser_version=__version__,
90                 )
91             return 0
92
93             for r in parsed.rows:
94                 dbmod.insert_axis_transaction(
95                     conn,
96                     document_id=doc_id,
97                     account_number=parsed.account_number,
98                     txn_date=r.txn_date,
99                     value_date=r.value_date or "",
100                     particulars=r.narration,
101                     debit=r.debit,
102                     credit=r.credit,
103                     balance=r.balance,
104                 )
105                 dbmod.record_parse(
106                     conn,
107                     doc_id,
108                     confidence=Confidence.RECONCILED.value,
109                     parser="AxisParser",
110                     parser_version=__version__,
111                 )
112             return len(parsed.rows)
113
114
115     # --- Credit card (no running balance; count only)
-----
116
117

```

```

118 @dataclass
119 class AxisCCTxn:
120     txn_date: str
121     particulars: str
122     merchant_category: str
123     amount: float
124     type: str # "Cr" or "Dr"
125
126
127 def _rows_from_cc_tables(tables: list) -> list[AxisCCTxn]:
128     """Build credit-card transactions from extracted tables. Pure: no PDF, no DB."""
129     rows: list[AxisCCTxn] = []
130     for table in tables or []:
131         for row in table or []:
132             if not row or len(row) < 9:
133                 continue
134             date_str = str(row[0]).strip() if row[0] else ""
135             if not _CC_DATE_RE.match(date_str):
136                 continue
137             amount_str = str(row[8]).strip() if row[8] else ""
138             is_cr = "Cr" in amount_str
139             amount = parse_amount(re.sub(r"^\d.", "", amount_str)) or 0.0
140             rows.append(
141                 AxisCCTxn(
142                     txn_date=date_str,
143                     particulars=str(row[2]).strip() if row[2] else "",
144                     merchant_category=str(row[7]).strip() if row[7] else "",
145                     amount=amount,
146                     type="Cr" if is_cr else "Dr",
147                 )
148             )
149     return rows
150
151
152 def parse_axis_cc(pdf_path: Path, password: str | None = None) -> tuple[str,
list[AxisCCTxn]]:
153     """Open an Axis credit-card statement and return (card_number, transactions)."""
154     card_number = "UNKNOWN"
155     rows: list[AxisCCTxn] = []
156     with pdfplumber.open(pdf_path, password=password) as pdf:
157         head = "\n".join((pdf.pages[i].extract_text() or "") for i in range(min(3,
len(pdf.pages))))
158         card_number = _first_match(head, _CARD_RES) or "UNKNOWN"
159         for page in pdf.pages:
160             rows.extend(_rows_from_cc_tables(page.extract_tables()))
161     return card_number, rows
162
163
164 @register_parser
165 class AxisCreditCardParser(StatementParser):
166     def can_handle(self, text: str, sender: str | None = None, subject: str | None =
None) -> bool:
167         return "Axis Bank" in text and "Credit Card Statement" in text
168
169     def parse_and_insert(
170         self, pdf_path: Path, conn: sqlite3.Connection, doc_id: int, password: str |
None = None
171     ) -> int:
172         try:
173             card_number, rows = parse_axis_cc(pdf_path, password)
174         except ParseError:
175             raise
176         except Exception as exc:
177             raise ParseError(
178                 f"failed to parse Axis CC statement {redact_filename(pdf_path)}"

```



```

179         ) from exc
180     for t in rows:
181         dbmod.insert_axis_cc_transaction(
182             conn,
183             document_id=doc_id,
184             account_number=card_number,
185             txn_date=t.txn_date,
186             particulars=t.particulars,
187             merchant_category=t.merchant_category,
188             amount=t.amount,
189             type=t.type,
190         )
191     # No running balance to reconcile; documents.status stays as the caller set it.
We still
192     # record provenance (with no confidence verdict).
193     dbmod.record_provenance(
194         conn,
195         document_id=doc_id,
196         source_type="pdf_statement",
197         parser="AxisCreditCardParser",
198         parser_version=__version__,
199         mapping_layer="deterministic",
200     )
201     return len(rows)
202

```

11. user (2026-06-10T16:29:28.755Z)

```

tests\parsers\test_axis.py:121:      ["05/02/2026", "x", "Swiggy", "", "", "", "", "Food",
"250.00 Dr"],
tests\parsers\test_axis.py:122:      ["06/02/2026", "x", "Refund", "", "", "", "", "Food",
"100.00 Cr"],
tests\parsers\test_axis.py:128:  assert rows[0].amount == 250.0 and rows[0].type == "Dr" and
rows[0].merchant_category == "Food"
tests\parsers\test_axis.py:129:  assert rows[1].amount == 100.0 and rows[1].type == "Cr"
tests\parsers\test_hdfc.py:221:  HDFC's own SUMMARY (Dr/Cr counts + opening/closing) and
reconcile fully.
tests\parsers\test_hdfc.py:236:      f"dr={report.debit_count_ok} cr={report.credit_count_ok}
"

```

12. user (2026-06-10T16:29:40.335Z)

.venv\Scripts\python.exe

13. user (2026-06-10T16:29:40.753Z)

65: # Credit-card-style rows (balance=None) still validate against the opening/closing bookends.

14. user (2026-06-10T16:29:59.473Z)

```

parse_amount('1,500.00 Dr') = 1500.0
parse_amount('1,500.00 Cr') = 1500.0
verified: True layer: L2
column_map: {'txn_date': 0, 'narration': 1, 'debit': 3, 'credit': 2, 'balance': 4}
  date=01-04-2026 debit=None credit=500.0 balance=1500.0 narr='UPI OUT merchant-a p'
  date=02-04-2026 debit=300.0 credit=None balance=1200.0 narr='NEFT IN salary-part '
  date=03-04-2026 debit=None credit=800.0 balance=2000.0 narr='UPI OUT merchant-b p'
reconcile_total under inversion (opening=1000, closing=2000 magnitudes): True
correct mapping reconciles: False breaks: 2

```

15. user (2026-06-10T16:30:40.299Z)

verified: True layer: L2 map: {'txn_date': 0, 'narration': 1, 'debit': 3, 'credit': 2,

'balance': 4}

16. user (2026-06-10T16:31:09.445Z)

Structured output provided successfully

17. assistant (2026-06-10T16:31:15.883Z)

Verification complete.